

# Technical Report: Neural Cellular Automata (NCA) for 2D-to-3D Procedural Generation and Bio-Inspired Natural Noise

## Executive Summary

Neural Cellular Automata (NCA) present a novel, paradigm-shifting methodology for introducing biologically inspired natural noise into procedural generation systems. Unlike classical mathematical noise algorithms—such as Perlin, Simplex, or Gaussian noise—which rely on deterministic global functions, NCAs generate spatial and volumetric structures through localized, emergent self-organization. By coupling neural perception with local update rules, NCAs exhibit extraordinary dynamic properties, including damage resilience, real-time structural self-repair, and continuous state-to-state morphing. This technical report details the elevation of NCAs from 2D spatial layouts to 3D volumetric representations, demonstrating how condition-driven transition training drastically enhances both generative quality and structural robustness.

## 1. Introduction & Problem Statement

**Problem Statement:** Exploration of Neural Cellular Automata (NCA) and its elevation from 2D spatial layouts to 3D volumetric representations.

**NCA & Natural Noise Generation:** Inspired by biological developmental processes and cellular systems like Conway's Game of Life, Neural Cellular Automata operate via local interactions to form complex global structures. Rather than synthesizing patterns using static mathematical equations, NCAs learn iterative local update rules that introduce dynamic, organic, and biologically authentic natural noise into procedural generation workflows.

**Key Applications:** This framework offers rich potential across multiple domain applications: synthesis of complex 2D and 3D architectural floor plans, smooth state-to-state morphing between distinct interior layouts, and robust object regeneration capable of recovering from severe physical damage injected during training.

**Code & Project Repository:** <https://github.com/oghritik/2D-NCA-to-3D-model>

## 2. Methodology & 2D-to-3D Pipeline

### 2.1 Categorical Base Image Generation & Preprocessing



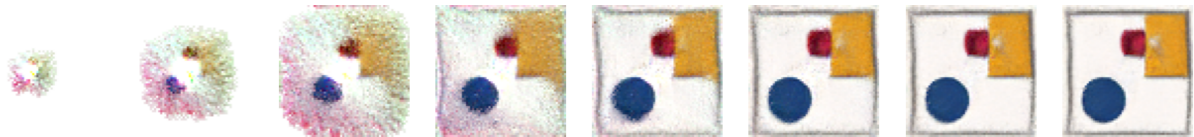
**Figure 1:** Base categorical layout (left), Gemini-enhanced floor plan (center), and target 64x64 rescaled matrix (right).

**Categorical Color Mapping:** Initial floor plans are defined using explicit semantic colors: **Black** represents perimeter/interior walls, **Yellow** denotes desks/tables, **Red** indicates seating/chairs, and **Blue** designates carpeted zones.

**Generative Enhancement:** The foundational layout is passed to Gemini to generate high-fidelity, visually rich interior assets while preserving semantic geometry.

**Rescaling & Normalization:** The enhanced layout is normalized down to a standardized 64x64 pixel grid, forming the exact ground-truth objective for NCA convergence.

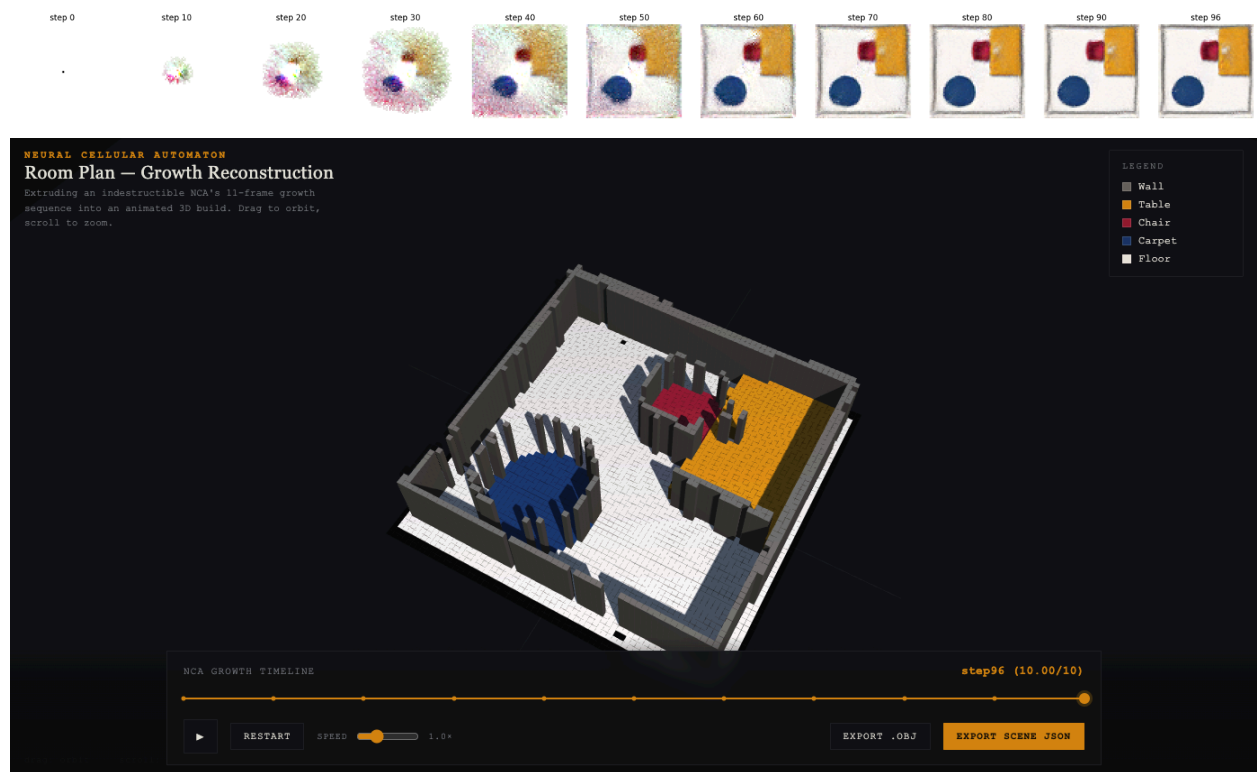
### 2.2 NCA Growth Dynamics



### 2.3 2D-to-3D Volumetric Elevation

To elevate 2D NCA growth patterns into full 3D volumetric models, we sample growth trajectories across key developmental milestones. Rather than relying purely on static 2D image frames, the model exports full high-dimensional cell states across intermediate steps. These state vectors encode dead/alive cell alpha channels, feature activations, and semantic category maps. Using custom WebGL/HTML rendering engines developed alongside Claude, these developmental trajectories are extruded into interactive 3D voxel fields with dynamic spatial height mapping, real-time orbit controls, and voxel extrusion timeline controls.

**Figure 2:** Developmental 11-frame growth trajectory from seed state to fully synthesized room layout, alongside its interactive 3D extruded voxel build.

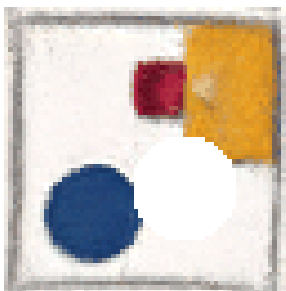


## 3. Model Architecture

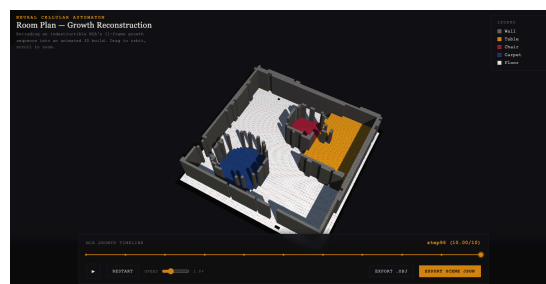
### 3.1 Single Layout Generation & Self-Healing

In the single-layout regime, the NCA is trained to grow from a single active seed pixel into a fully realized target structure. The network learns homeostatic self-maintenance through **sample pool training with damage injection**. During training, random circular cutouts are erased from intermediate cell grids in the pool, forcing the network to learn localized repair mechanisms alongside growth dynamics.

NCA:

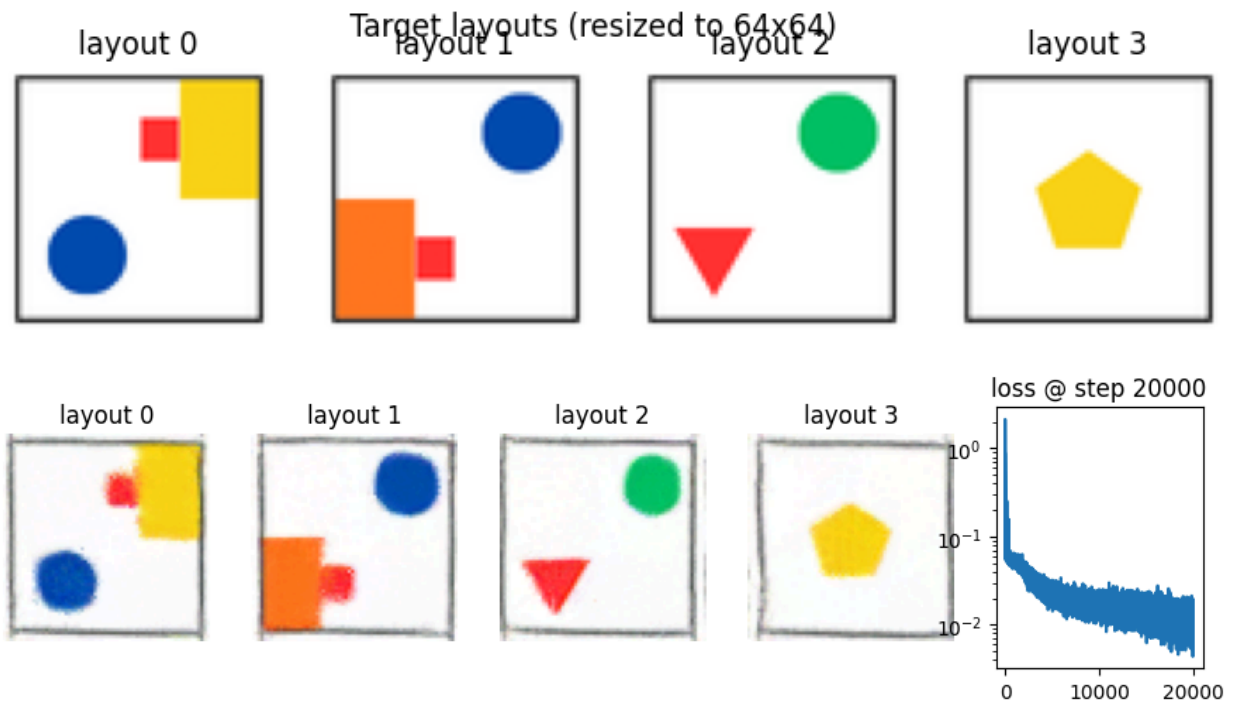


Converted 3D model:

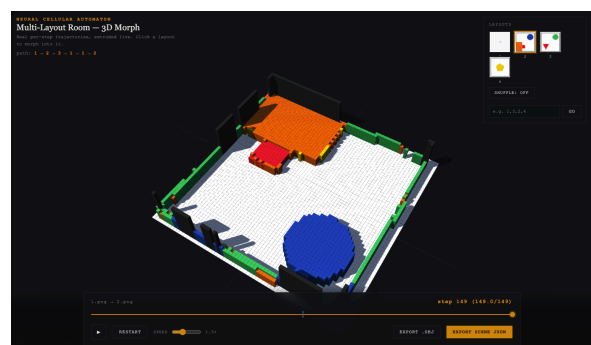
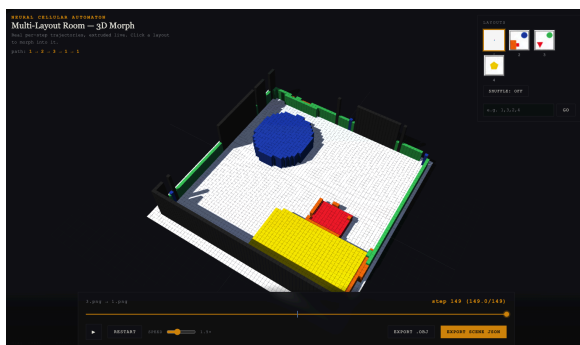


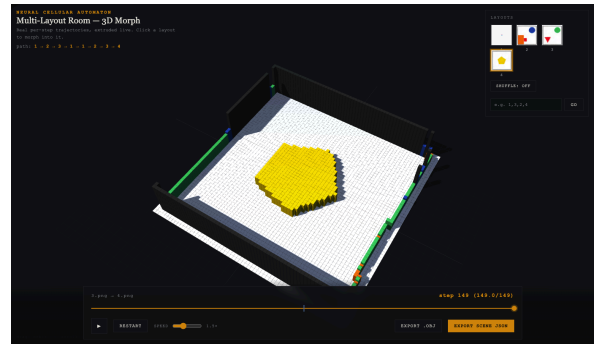
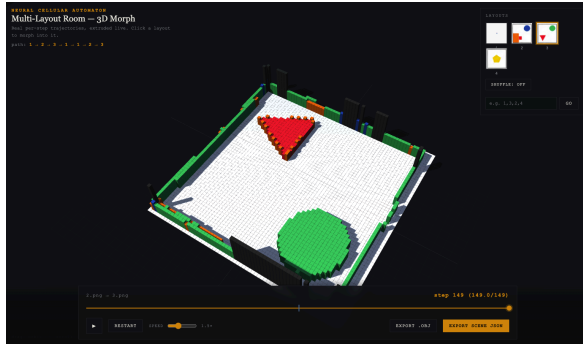
### 3.2 Multi-Layout Generation Without Transition Training

When expanding the NCA to learn four independent interior layouts simultaneously, the network is conditioned via standard target layout vectors. While the model succeeds in growing each layout from an initial seed block, it **fails to transition** smoothly between active states. Furthermore, under severe damage injection, the absence of inter-state regularization causes the model to decay or completely erase structural features upon perturbation.



All four 3D converted layout:





## Baseline NCA Architecture

Perception Layer (Fixed Sobel filters + Identity filter across 16 state channels -> 48 dimensions per cell)

↓  
Conv2d(48 → 128, kernel=1×1) ← learned layer 1

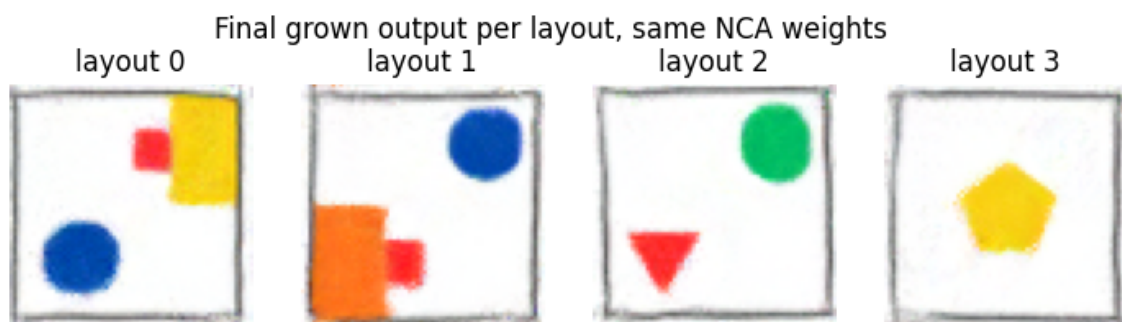
↓  
ReLU

↓  
Conv2d(128 → 16, kernel=1×1) ← learned layer 2 (no bias)

## 3.3 Multi-Layout Generation With Transition Training

To overcome transition failures, we incorporate explicit layout-shuffling and state-morphing steps into the training loop. By forcing the NCA to dynamically morph from one target layout state to another, transition training acts as a powerful global regularizer. The resulting model not only morphs smoothly across all four distinct configurations, but its self-healing resilience is fundamentally improved—rapidly reconstructing destroyed room sections without losing structural integrity.

In this I have also used a different method to convert the 2d model into 3d model using python and html. Earlier only 11 frames were used to construct the 3d model but now nca code set itself returns a list of nca states which include intermediate states and dead/alive information of a nca cell. This Benefits the 3d model construction in creating more accurate model. The model has better colour mapping method.

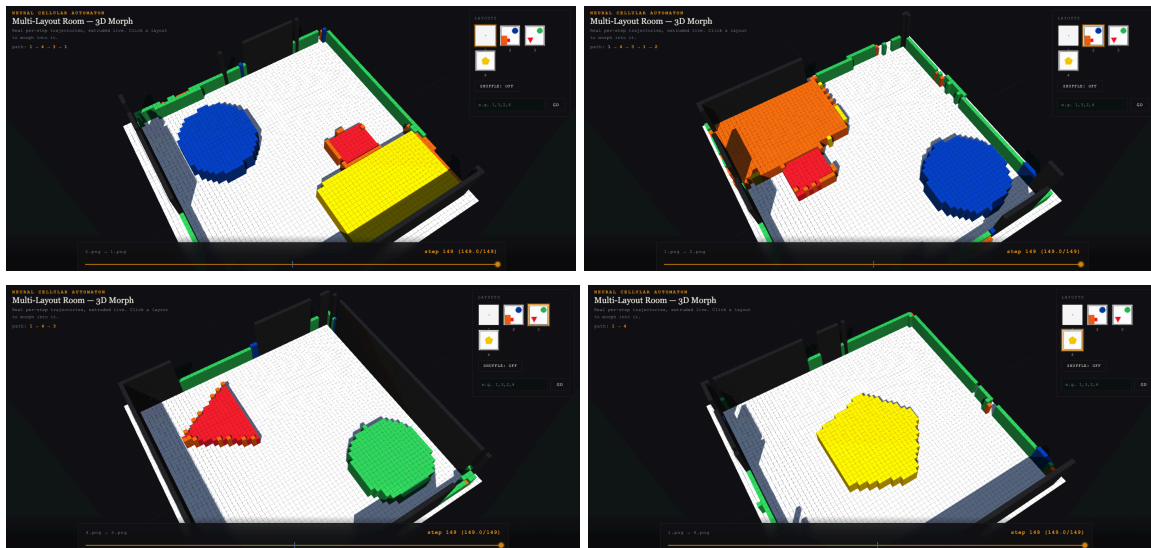


running | brush=4 | layout 0: 1.png  
 Growing layout 0: 4/200



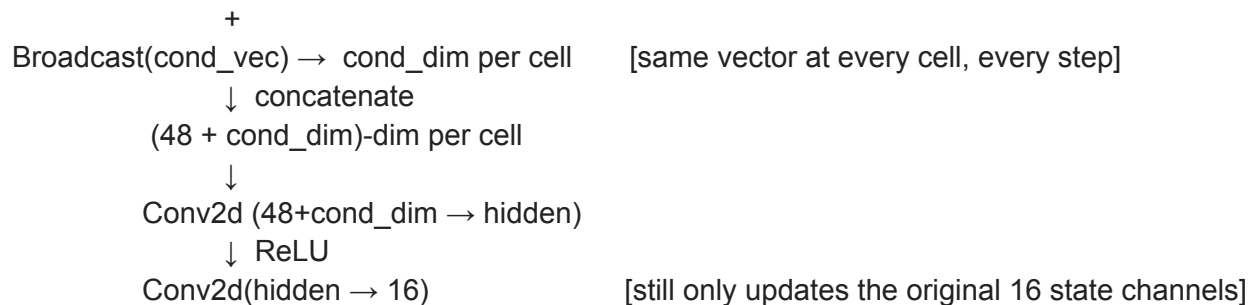
### 3D Conversion:

Models look the same but they have the ability to go from one state to another easily.



### Conditioned Multi-Layout Architecture

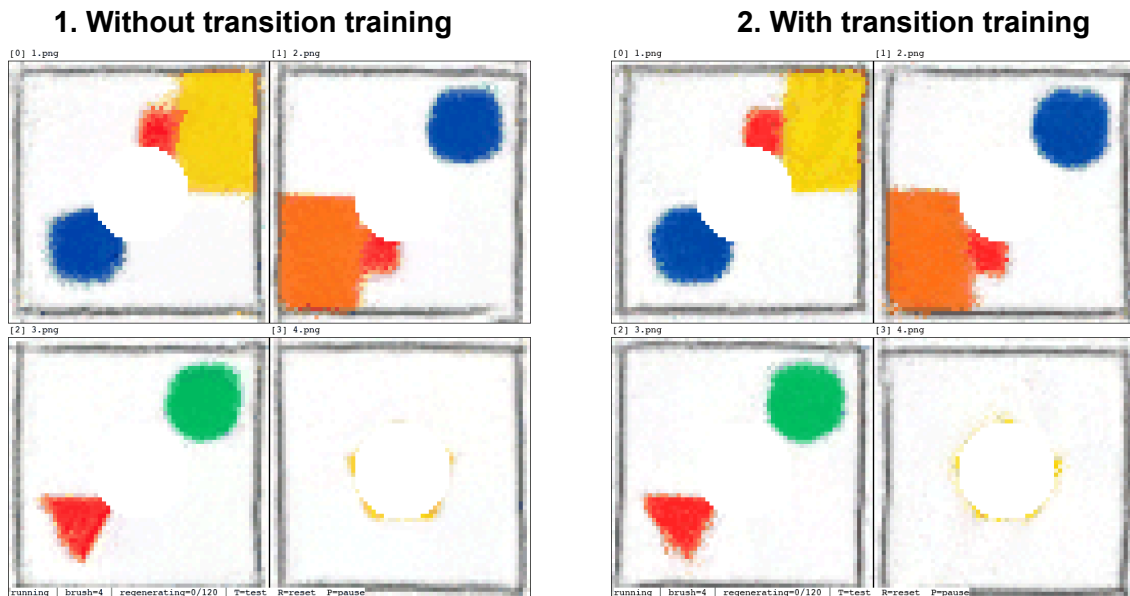
Perception Layer (State Perception: 48-dim per cell) + Conditioning Vector Broadcast (cond\_dim per cell)



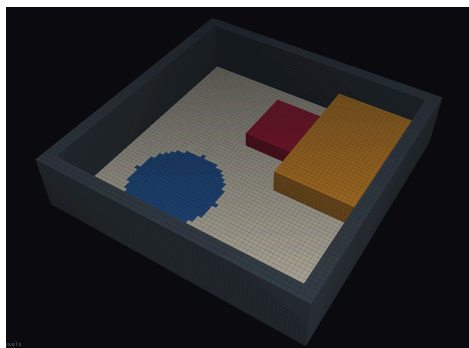
## 4. Experimental Results & Structural Regeneration

**Ablation Observations:** When trained purely on static target reconstruction without state morphing, damage injection frequently led to catastrophic fading or total structural collapse. Conversely, introducing transition conditioning enabled robust, biological-grade self-repair across all target states.

But when the model was trained for 4 distinct model regeneration and also for transition, the models ability to regenerate was very much improved.



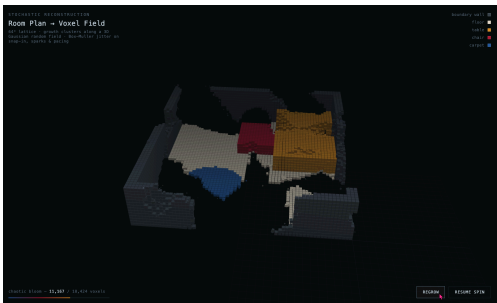
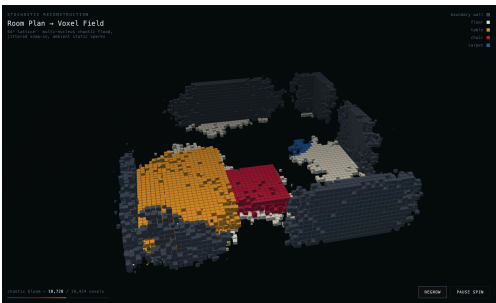
## 5. Comparative Analysis: Bio-Inspired Noise vs. Classical Procedural Noise



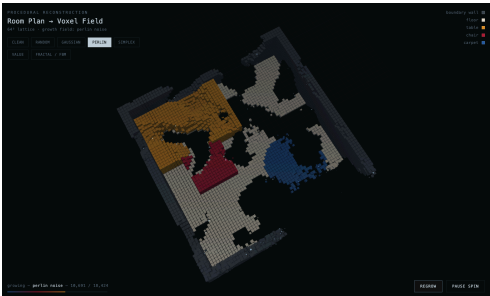
Clean non-noisy generation

To evaluate the generative characteristics of Neural Cellular Automata, we contrast NCA-synthesized volumetric fields against classical procedural generation techniques, including direct Claude 3D model prompting as well as mathematical noise fields (Perlin, Simplex, Gaussian, Value, and Fractal/FBM noise).

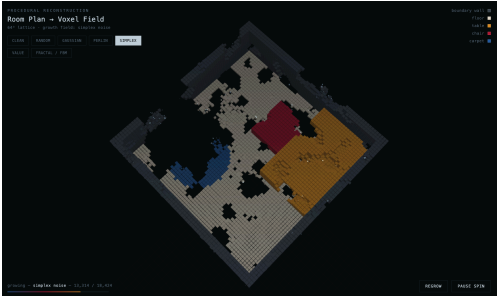
- 1. The model seems very clean with no noise and jitter.
- 2. Gave the exact model which was expected.



Random Noise



Gaussian Noise



Perlin Noise

Simplex noise, etc.

Neural Cellular Automata derive their structural properties from localized cellular interactions, mirroring biological morphogenesis. While classical noise functions offer exact, analytical evaluation without training overhead, they produce static, non-adaptive spatial distributions. NCAs trade initial training complexity for unprecedented dynamic adaptability, self-repair, and organic spatial emergence.

| Characteristic  | NCA                                 | Non-NCA                                                                                  |
|-----------------|-------------------------------------|------------------------------------------------------------------------------------------|
| Nature of Noise | Natural noise                       | Synthetic, continuous mathematical distributions (Perlin, Simplex, Gaussian, Value, FBM) |
| Dependency      | High dependency on training quality | Independent (no training required)                                                       |

| Characteristic | NCA                             | Non-NCA                   |
|----------------|---------------------------------|---------------------------|
| Complexity     | High (inspired by Game of Life) | Lower (direct generation) |
| Visual Quality | Chaotic/Organic                 | Clean/Precise/No jitter   |

## 6. Conclusion & Limitations

1. **2D-to-3D Volumetric Pipeline:** Successfully implemented an automated trajectory-extraction pipeline elevating 2D NCA developmental states into interactive 3D voxel models.
2. **Regularizing Effect of Transition Training:** Demonstrated that training multi-layout NCAs on dynamic state morphing drastically boosts structural stability and damage regeneration performance.
3. **Bio-Inspired Procedural Noise:** Validated NCAs as a superior paradigm for generating organic, biologically authentic natural noise in complex procedural design systems

### 6.1 Current Drawbacks & Future Directions

- High-frequency persistent boundary noise during steady-state equilibrium remains an inherent artifact of local cellular updates.
- Current 3D extrusion relies heavily on contiguous floor anchor planes; unanchored floating geometries present structural connectivity challenges that warrant dedicated 3D NCA kernel extensions.